Sora Kim

Independent Researcher, Pharmaceutical GMP Quality Control

sorakim.lab@gmail.com +82 10-8119-7439 Seoul, South Korea

sorakim-lab.github.io sorakim-lab ORCID: 0009-0003-0856-4193

SSRN: ssrn.com/author=10648616

Research Focus

Responsibility is produced, not assigned. My research program—**Accountability-Aware AI**—investigates how regulated organizations structure responsibility before failures occur, and how AI integration reshapes these accountability infrastructures. Drawing on 4 years of embedded practice in pharmaceutical Good Manufacturing Practice (GMP) quality control, where I have deployed 9 production tools serving a ~50-person quality team, I combine ethnographic, qualitative, computational, and design-science methods to examine accountability both as a structural property of regulated work and as a design challenge for AI-integrated systems. This approach sits at the intersection of HCI, STS, and organizational technology adoption.

Education

Chungbuk National University *Cheongju, South Korea*

B.A. in English Literature and Linguistics March 2015 – August 2022

- **Top-Ranked Admission**, Humanities & Social Sciences Division, 2015 (entrance examination).
- **Full Merit Scholarship** (tuition & fees, 8 semesters; merit-based).

Industry Research Experience

GMP-Regulated Pharmaceutical Manufacturing Site *Seoul, South Korea*

Quality Control / Quality Systems September 2022 – Present

- Embedded practitioner-researcher in regulated biomanufacturing quality operations (4 years as of application cycle).
- Conducted sustained ethnographic observation of Standard Operating Procedure (SOP) use, regulatory deviation handling, Corrective and Preventive Action (CAPA) processes, and regulatory audit workflows.
- Designed and deployed 9 productivity tools serving Quality Control (QC) operations (see Deployed Systems).
- Designed and delivered a differentiated AI training program for the QC team (managers: tool-building; operators: AI-assisted authoring), establishing the team's first structured reference for AI in GMP workflows; observations from this program inform ongoing research on AI

integration in regulated environments.
- Observational data and fieldwork form the empirical foundation of the research program below.

Deployed Systems *(Empirical Substrate for HCI Papers)*

Production tools designed and deployed at a GMP-regulated pharmaceutical manufacturing site. All systems operate as unvalidated productivity layers external to validated GMP systems, with team lead awareness and support. These systems are not peripheral artifacts but the empirical substrate of the HCI papers below—each tool serves as a site of ethnographic observation, behavioral data collection, and design iteration, and as a concrete test case for how AI can be integrated into regulated work without eroding accountability.

- **QC Stability Testing Dashboard** — Deadline-tracking interface for stability testing programs; team-wide adoption, ongoing operation.
- **SOP Retrieval System** — Semantic search tool for GMP procedural documents; basis for *From Retrieval to Reasoning* (see Publications).
- **Document Navigation Tool (Traceability Layer)** — Cross-document linking interface with metadata mapping across regulated quality documentation; empirical base for *When Documents Don't Connect* (traceability component).
- **Navigation Behavior Tracker** — Behavioral tracking layer capturing navigation paths, hesitation, and cross-reference jumps in regulated document work; empirical base for *When Documents Don't Connect* (loopback component).
- **Review Dashboard** — Prioritized document review interface surfacing procedural friction before formal deviation; basis for *Friction Before Failure*.
- **Root Cause Analysis (RCA) Scaffolder** — Investigative scaffold that preserves analyst judgment while structuring causal reasoning; basis for *Keeping Comparison Open*.
- **Veeva Vault Change Control Assistant** — Workflow support for electronic change control submissions.
- **Reagent Inventory & Expiry Management Tool** — Automated expiry tracking and inventory alerts for QC reagents.
- **Document Auto-Review AI** — Format and compliance pre-check for procedural documents.

Publications

STS & Organizational Theory — Under Review

A four-paper lifecycle theory of accountability in regulated organizations: pre-action structuring (AAC), procedural document infrastructure (CF), investigative convergence (RC), and AI-accelerated closure (PAC). The lifecycle articulates how responsibility is produced structurally at each stage, with implications for AI integration in regulated work.

Kim, S. (2026). *Anticipated Accountability Convergence: How Organizations Structure Responsi-*

bility Before Action. Under review at **Academy of Management Review** (AMR-2026-0516). SSRN: 6371980

Kim, S. (2026). *Cognitive Friction: Procedural Documents as Accountability Infrastructure in Regulated Organizations.* Under review at **Organization** (ORG-26-0241). SSRN: 6391278

Kim, S. (2026). *Responsibility Convergence: How Investigative Systems Stabilize Attribution.* Under review at **Organization Science** (ORSC-MS-2026-22511). SSRN: 6372301

Kim, S. (2026). *Premature Accountability Convergence: How Technology Accelerates Responsibility Stabilization in Organizations.* Under review at **Information and Organization** (INFORG-D-26-00222). SSRN: 6391379

HCI & Design — In Preparation

5 HCI papers on judgment-preserving design in regulated work, each grounded in a deployed production tool (see Deployed Systems). Together they articulate a four-layer framework—accessibility, structure, coordination, reasoning—for accountability infrastructure design in AI-integrated regulated environments.

Kim, S. (2026). *Accountability Infrastructure Design: A Four-Layer Framework for Judgment-Preserving Systems in Regulated Work.* In preparation for submission to **ACM Transactions on Computer-Human Interaction (TOCHI)**.

Kim, S. (2026). *Keeping Comparison Open: Interface Scaffolding Against Premature Accountability Convergence in Regulated Investigations.* In preparation for submission to **CSCW 2027**.

Kim, S. (2026). *Friction Before Failure: Making Review Burden Visible in Regulated Document Work.* Working paper (SSRN).

Kim, S. (2026). *From Retrieval to Reasoning: How SOP Accessibility Shapes Interpretation in Regulated Work.* Working paper (SSRN).

Kim, S. (2026). *When Documents Don't Connect: Traceability Burden and Loopback Behavior in GMP Document Navigation.* Working paper (SSRN).

Computational — Working Papers

A computational research program operationalizing accountability across four signal layers (detection, graph structure, attribution, distribution), applied to FDA Warning Letters and synthesized in the umbrella paper. These analyses provide external validation of the lifecycle theory through large-scale regulatory text.

Kim, S. (2026). *Responsibility as Multi-Layer Signal: A Computational Framework for FDA Warning Letters.* Working paper (SSRN). [Umbrella]

Kim, S. (2026). *Premature Convergence Detection in Regulatory Investigative Texts.* Working paper + GitHub: sorakim-lab/premature-convergence-detection.

Kim, S. (2026). *Accountability Graph Extraction from Regulatory Texts.* Working paper (SSRN).

Kim, S. (2026). *Responsibility Attribution Prediction: A Classification Approach to FDA Warning Letters.* Working paper (SSRN).

Kim, S. (2026). *Responsibility Distribution Metric: An Entropy-Centrality Approach to Account-*

ability Structure. Working paper (SSRN).

Conferences

- Kim, S. (2026). *Premature Accountability Convergence: Inscription, Closure Pressure, and the Stabilization of Responsibility.* Abstract submitted to **4S (Society for Social Studies of Science) Annual Meeting**, Toronto, 2026 (Panel 74).

Research Corpus & Data Infrastructure

- **FDA Warning Letters Corpus** — N=50 letters, 160 violations (2016–2025). Used across STS, organizational theory, and computational papers.
- **Cognitive Friction Corpus** — 400 naturally occurring sentences from regulated procedural documents, manually annotated at sentence level.
- **Field Observation Archive** — 4 years of notes on SOP use, deviation handling, and CAPA in a GMP-regulated site.

Methods & Technical Skills

Research Methods: Ethnographic fieldwork; document & corpus analysis; within-subject experimental design; think-aloud protocols; behavioral logging; qualitative coding; annotation protocol design; causal reasoning analysis.

Computational: Python, R, Stata; NLP & text processing (TF-IDF, sentence-level annotation, classification); Streamlit (deployment); LaTeX; HTML/CSS; Git/GitHub.

Regulated-Environment Systems: Veeva Vault QMS; GMP documentation systems; electronic change control; deviation/CAPA workflows.

Languages

Korean (native) · English (fluent) · Japanese (fluent)